Prepared for: M/s. Masa Iron Products Factory Company LLC
Quotation No: QEXP12878
6000 kW DM (HE) Panel with 15 Ton Steel Frame Furnaces

Inductotherm (India) Pvt. Ltd.
Plot No. SM – 6, Road No. 11,
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BOL Village, Sanand,
Ahmedabad - 382 170, India
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M/s. Masa Iron Products Factory Company LLC
3rd Industrial City, Dammam
Saudi Arabia

Kind Attn.: Mr Khalid Al Abdullah
Mob.: +966 561188855
Email.: info@massa-steel.com

Sub: Your Requirement of Induction Melting Furnace

Dear Sir,

This is in reference to your discussions with undersigned in connection with your requirement of Medium Frequency Induction Melting Furnace. We really appreciate your interest in Inductotherm Induction Melting Furnace.

As per your discussion, we are pleased to enclose herewith following quotation for your perusal:

- **Quotation for** ONE [1] 6000 KW/200 HZ VIP DUAL MODE HIGH EFFICIENCY WITH TWO [2] 15000 KG STEEL FRAME FURNACES.

We are also enclosing herewith technical specification sheet, scope of supply, standard terms & conditions.

Hope our offer is in line with your requirement. If you need any further information/assistance from our side, please feel free to contact undersigned or our **Mr. Sunil Pandita, Asst. General Manager (Sales), (Cell # 09737377734).**

Thanking you,

Sincerely,

PRASEN MEHTA
ASST. MANAGER (SALES)
CELL: 09374954573
Encl: Quotation consists of price sheet, technical specification, bulletins, international terms & conditions (ITAC-03).

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THE COMPANY

1. INDUCTOTHERM CORPORATION

Inductotherm Group leads the industry in the development and manufacturing of advanced technologies, products and systems for the heat-driven transformation of metals and specialty materials. You'll find us wherever our customers need to melt, heat treat, forge, roll, cut and weld.

Inductotherm Group is structured to reach across the globe and organized to provide localized manufacturing, engineering, service and support in every region of the world. We do it through 40 remarkable full-service companies providing 50 individual product lines organized into 10 global brands. Each product is designed for superior and consistent performance and offered around the world by a highly developed technology transfer and manufacturing system.

Across six decades, Inductotherm Group has grown, adapted and evolved. The result: A broad offering of products and services to meet the needs of customers across every industry.

Inductotherm Group has supplied more than 35000 Induction Melting Furnaces in 131 different countries and operates from **Argentina, Australia, Belgium, Brazil, Canada, China, France, Germany, India, Indonesia, Japan, Korea, Mexico, Russia, Spain, Turkey, Taiwan, U.K. & USA.**

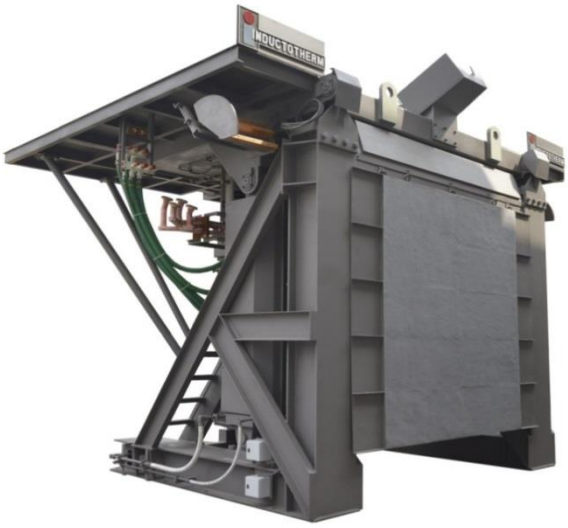
2. INDUCTOTHERM GROUP INDIA, AHMEDABAD

Inductotherm (India) Pvt. Ltd. an Inductotherm Group company leads the industry in the development and manufacturing of advanced technologies, products and systems for the heat-driven transformation of metals. Company has been actively working and supporting the metal industry with products and services since 1982.

With the state-of-the-art manufacturing plant in Sanand, spread across 100,000 sqm is fully equipped with the latest machines to meet the thermal challenges of the industry for varied applications. Since beginning, Inductotherm has worked to build a reputation for innovation, exceptional engineering and unequalled standards of quality, offering the most reliable solutions and has emerged as a true leader in induction technology with more than 5000 equipment's running successfully.

Company excels the leadership position in induction technology for Melting, Heating, Welding, Vacuum melting, Vacuum heating and Long products.

3. THE FURNACE DIVISION

Furnace Division of Inductotherm (India) builds Induction Melting, holding (both coreless and channel type) and Pouring Systems (both heated and unheated) for virtually all metals including iron, steel, copper, aluminium, zinc, precious metals etc. The melting division's product line also includes a full range of charging systems, melt shop computer control systems and automated robotic melt shop systems. We not only installed more than **5000** Induction Melting furnaces in India but have also successfully carved out a position as a market leader in all the segments in India and are servicing more than **5500** Inductotherm Installations including imported ones. This is out of estimated population of **12000** Medium Frequency Induction Furnaces all over India.	

Some of our achievements detailed below will give you an idea about our capabilities to serve metal industries:

- 24000 kW 2 sets VIP® Power-Trak® with 4 x 50 Ton furnaces and 22000 kW 4 sets VIP® Power-Trak® with 8 x 50 Ton furnaces for the world's largest steel melt shop using induction furnaces at BSRM, Bangladesh.
- 16000 kW VIP® Power-Trak HE power supply system supplied to JINDAL STAINLESS, HISAR with 45 Ton STEEL FRAME FURNACE.
- 7500 kW VIP® Tri-Trak™ (16000 kW Rectifier) induction melting system supplied to AIA ENGINEERING LTD, AHMEDABAD.
- 12000 kW VIP® Dual-Trak® induction melting system supplied to Jindal Saw Gulf L.L.C – UAE.
- 55 Ton channel-type furnace supplied to Sakthi Auto Components, Coimbatore.
- 15000 kW with 2 x 40 Ton steel frame furnace supplied to Welspun Steel Ltd, Anjar.
- 3000 Kg. Multi Pour with vision control supplied to BRAKES INDIA, NAIDUPET.
- 14000 kW bar re-heater prior to rolling supplied to Kamachi Power & Steels, Chennai.
- 6200 kW modular billet heater for Eumoco 10000 Ton press supplied to Bharat Forge Ltd., Pune.
- 5600 kW modular billet heater for SMS 12500 Ton press supplied to Ramkrishna Forgings Ltd.- Jamshedpur.
- 3800 kW modular billet heater for Sumitomo 4500 Ton press supplied to Honda Cars India Ltd., Alwar.
- 7500 kW induction long bar heater for Hatebur HM 75 XL hot former supplied to Rolex Rings – Rajkot.
- 1800 kW IGBT booster heater prior to rolling supplied to Kalyani Group, Pune.
- 550 kW Induction hardening equipment for Rear Axles Hardening supplied to GNA Axles Ltd. – Mehtiana.
- 1400 kW end bar heater for Ficep vertical press supplied to GNA Axles Ltd., Mehtiana.
- State of art HAZControl™ Technology HCT 400 welder producing high quality precision steel tubes supplied to Good Luck Industries, Ghaziabad.
- VIP® 800 kW Smart Anneal systems, latest in API line pipe seam normalizing supplied to Saudi Steel Pipes, Dammam.
- CFD 1000 kW solid-state dual welder for the manufacture of API Line pipes supplied to Maharashtra Seamless - Raigad & Jindal (India), Howrah.
- 3 x 60 MT Zinc cathode melting channel furnaces supplied to Hindustan Zinc Ltd., Chittorgarh & Haridwar.

TECHNOLOGY

The following is summary of features which will be incorporated in any equipment.

Features highlighted in RED are unique to Inductotherm.

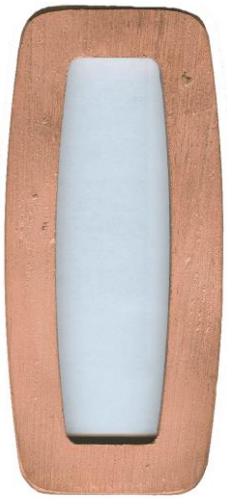
Features highlighted in PURPLE are designed to enhance and improve refractory lining life.

Furnace Design: Inductotherm's heavy steel frame furnace design incorporates the following features:

1) This rugged construction enables Inductotherm to build the most rigid furnaces available.

2) Inductotherm provides Iron shunts which cover up to 60% of the outside of the coil to ensure adequate control of the magnetic field and direct mechanical support of the coil and better electrical efficiency

3) Inductotherm's coil is manufactured from heavy wall copper tubing to provide strength and large water paths to ensure adequate cooling to the back of the refractory lining.



4) Inductotherm has been building high efficiency “studded” coils since the 1960's. Phosphor bronze coil studs are used at all times.

5) Coils are manufactured with the minimum of joints and all joints are made using a beveled, sleeved joint that is brazed and identified.

6) Inductotherm’s coils incorporate low loss stainless steel cooling coils above and below the power coil to ensure adequate cooling to the back of the refractory lining through the full height. Low conductivity stainless steel is used to construct the cooling coils to reduce the amount of energy induced in to cooling section thereby increasing the energy available for melting and reducing the losses to cooling system.

7) Spacers placed between the coil turns in 16 positions around the circumference of the coil ensure rigidity of the coil in the vertical and provide a means for the coil to “breath”. The coils ability to breath ensures the free passage of moisture from the refractory lining during the sinter cycle.

8) Inductotherm’s power connect is positioned halfway up the furnace. This eliminates the need for a pit in the foundation behind the furnace, minimizes lead movement and removes the leads from the area directly under the back of the furnace.

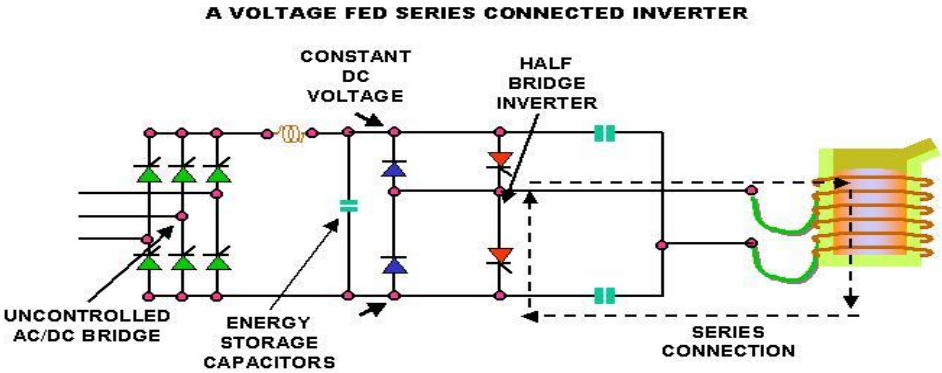
9) The tilt cylinders are fitted with flow restrictors. If a hydraulic hose or line bursts the furnace will drop at a controlled rate.

10) The furnace top and bottom refractories are manufactured as pre cast, pre fired shapes incorporating stainless steel fibers for additional strength and durability in preference to castable refractory applied on site.

11) The furnace weighing system is based on load cells (Optional). To enhance the accuracy and simplify the installation on the foundations Inductotherm supplies the load cells pre mounted in a structural steel frame which is located under the furnace.

12) Inductotherm's coils are insulated with a proprietary material called InductoFlex. The use of this material allows Inductotherm to operate with elevated coil voltages. High coil voltages mean lower currents for the same kilowatt draw resulting in lower electrical loses. It also means higher voltages are induced into the charge which enhances the "contact" within the charge.

VIP POWER-TRAK power supply: The VIP Power Trak is based on Inductotherm's time proven voltage fed inverter power circuit. Over 36000 of these have been supplied worldwide since 1976 and more than 5,000 of these are rated at 5,000 kW and higher.



1) This Power -Trak operates with an uncontrolled AC/DC rectifier. An uncontrolled AC/DC rectifier ensures that the line power factor Cos Ø will always be between 0.98 and above at ALL levels of power draw.

2) Full Power from start to finish ensuring shorter tap to tap time.

3) The efficiency of the power supply is quoted as 98%.

4) The SCR's in this system will be operated at 50% of their manufacturer's KVA rating and selected for a optimum thermal junction temperature.

5) The combination of the high power factor and high efficiency ensure Inductotherm's power supplies have the lowest KVA requirement.

6) All Inductotherm's power supplies are built into steel enclosures. This ensures all the components, interconnecting cables and bus bars are installed in the factory and power supply is tested in-house prior to delivery. This design enables Inductotherm to simplify the installation of the equipment and shorten the onsite installation and commissioning period. This construction ensures the power supply sits on a flat floor and does not require any trenches.

7) The nominal operating frequency will be 200 Hz which gives the right balance on stirring and lesser power consumption and helps in melting light scrap in proper manner

MISCELLANEOUS FEATURES:

Inductotherm's designs incorporate a number of features which provide benefits to the customer.

1) FOUNDATIONS: All Inductotherm's equipment is design to mount of a flat concrete floor.

The power supply does not require any trenches as all connection are overhead.

An Inductotherm furnace with the power connections in place can fully tilt whilst standing on a flat concrete floor. Unlike our competitors who have the power connection at the bottom rear of the furnace, it does NOT require a pit behind it for the power leads.

2) The installation cost is kept to a minimum as each unit is pre wired and pre piped before delivery to site.

3) ISO 9001: Inductotherm's equipment is ALL built and tested in the same ISO 9001 approved factory.

TECHNICAL SPECIFICATIONS

6000 KW/200 Hz VIP DUAL MODE HIGH EFFICIENCY
(6 PHASE – 12 PULSE)

A.	**APPLICATION REQUIREMENTS**	
1.	Alloy to be melted	STEEL
2.	Melt temperature	1600°C
B.	**CHARACTERISTICS OF RECOMMENDED POWER UNIT**	
1.	Rated KW	**6000 KW**
2.	Converter KW	**6000 KW**
3.	Nominal furnace frequency	200 Hz
4.	Line power factor	0.97 and above
5.	KVA required at input of VIP Power Trak-R	**6380** KVA on load
6.	Melt rate at **6000 KW ****	**12310** Kg/hour – **Steel**
7.	Power connection	575 V, 6 Phase, 50 Hz
C.	**CHARACTERISTICS OF RECOMMENDED MELTING FURNACE**	
1.	Nominal capacity **(Steel Capacity)**	**15000 KG**
2.	Style of furnace	STEEL FRAME
3.	Pouring mechanism	Hydraulic tilt
4.	Furnace lining (Recommended - to be provided by the customer)	Silica

** The above melt rate is based on a nominal furnace size for second heat when lining is hot, charge is dense and bus runs proper. The voltage should be steady within allowable range. Cooling water should be as per our specification. Melt rates will be for the weight of charge and does not include time for initial charging, pouring, superheating, deslagging or chemical analysis. Please note that slag consumes nearly double the power.

PRICING FOR MELTING SYSTEM

6000 KW VIP DUAL MODE HIGH EFFICIENCY FURNACE SYSTEM		
Sr.	Quantity	Description
1.	One	**6000 KW, 12 Pulse, VIP POWER TRAK** Power & Control System comprising of - Power input section - Converter section rated for 6000 KW - Inbuilt DC capacitor in filter section - **New efficient** Voltage fed Inverter section rated for 6000 KW - Inbuilt AC tank capacitor section - Control and monitor section - husky copper bus bars within panel - Internal distilled water circulating system with standby pump and motor - 2nos furnace selector switches with suitable electrical interlocks - Hydraulic power supply unit with standby pump and motor - Remote control console
2.	One	**500 KW SINTERING Power & Control System comprising of** - DC switch - Inbuilt DC capacitor in filter section - Voltage fed Inverter section rated for 500 KW - Inbuilt AC tank capacitor section - Control and monitor section - husky copper bus bars within panel - furnace selector switches with suitable electrical interlocks - Remote control console
3.	Two	**15,000 KG STEEL FRAME FURNACE** comprising of: - - hydraulic tilting arrangement, - set of water-cooled leads - one standard set of Interconnecting copper tubing
4.	One	I-Optimizer Plus
5.	15 Days	Field Service for start-up commissioning
6.	2 sets	Approved Civil water system and foundation drawings.
7.	2 sets	Drawings & Operation Manuals.
PRICE: SR. NO. 1 TO 7 [FOB MUMBAI]		**USD 545,000**
AUXILIARIES		
8.	One	7475 KVA Water Cooled Furnace Transformer.
9.	One set	Transformer to Panel interconnecting arrangements
10.	One	FRP Cooling Tower. (Induced draft type)
11.	One set	Raw water, soft water and emergency pump
12.	One	Coil Cooling Heat Exchanger
13.	One	MCC Panel
14.	One	DM Plant
PRICE: SR. NO. 8 TO 14 [FOB MUMBAI]		**USD 205,000**

Delivery will be within Six [6] to Eight [8] months. Other terms and conditions are as per the enclosed international terms and conditions (Bulletin No. ITAC-03).

For INDUCTOTHERM (INDIA) PVT. LTD.

PRASEN MEHTA

ASST. MANAGER (SALES)

Your equipment will be made with use of sustainable Solar Power

SCOPE OF SUPPLY

A. POWER UNIT

I. ELECTRICAL PANEL

1. **CABINET**

Metal cabinet, duly painted fitted with panel doors which are gasketed and equipped with locks. In addition, micro switches are provided which illuminate a lamp on the monitor board and shut off power to prevent injury to personnel when the lift off panel or doors are opened.

Power connections are easily made through the top and water connections are through the side of the cabinet.

2. **RECTIFIER SECTION WITH FILTER**

a) High power SCRs with snubbers for rectification.

This rectifier is designed to minimize line harmonics compared to phase-controlled rectifier.

Instantaneous trip circuit facility is provided for the protection of the rectifier.

b) Fast acting semi-conductor fuses.

c) One No. of Air core encapsulated current limiting reactors.

d) DC capacitors located in capacitor section.

This design of converter and filter section reduces losses compared to iron core current limit reactor(s) and helps to provide constant DC voltage to the voltage fed inverter.

This design helps to achieve the conversion efficiency not less than 96%.

SCOPE OF SUPPLY - PAGE 2

3. **INVERTER SECTION**

This section contains inverter panel containing high power inverter SCRs with snubbers, anti-parallel diodes and DI/DT reactors. This helps to provide full power throughout the melt cycle.

4. **CAPACITOR SECTION**

a) This section contains all the required DC filters and medium frequency AC capacitors.

b) One pressure switch, installed in each capacitor.

c) One indicating lamp located on the monitor board to notify the operator when the capacitor pressure switch has been actuated.

5. **GROUND/METAL LEAK DETECTOR**

One sensing ground/metal leak detector ready to sense and indicate any ground and metal leak. Consisting of indicating lamp, millimeter, probe disconnect switch to disconnect the probe from the power supply.

6. **CONTROL AND MONITOR SYSTEM**

Enhanced Z-Control Board

Enhanced Z-Control Board (EZCB) is a next step in developing integrated control board with expanded diagnostic. Its design is based on ACCB, ZC-Control Board and Z-Control Board and uses the same principle: controls the delay time between inverter current zero crossing and the next SCR firing pulse depending on the feedback signals. The same new technology from ZDCB and ACCB is used on the EZCB: one FPGA (Field Programmable Gate Array) combines all-digital, logic and time setting/delay functions which facilitates setup and serviceability, adds more flexibility and functionality, and improves reliability. More diagnostic capabilities with priority decoder based on accumulated knowledge was added to FPGA and its size was doubled. Surface mounted technology (SMT) was used in the design to accommodate 144 pin FPGA IC.

The EZCB was designed as one control board in the unit and incorporates all sensors, actuators, indicators (two digit seven segment display) and firing. The main functions included in EZCB are:

- VIP Control (similar to ZDCB with some modifications)
- Soft Start ACI (similar to ACCB with small modifications)
- Four potential sensors (inverter voltage, DC Bus voltage, capacitor voltage and furnace voltage).
- Diagnostics (status, limit and alarm indicator)
- Inverter SCR Firing Module
- ACI Firing Module
- CLR SCR Firing Module (additional protection from over voltage)
- KWMR
- Lost Phase Detector
- Wrong Phase Rotation Detector
- Short Coil Detector
- Short Secondary Capacitor Detector
- DC Balance Monitor
- AC Current Limit Circuit with two sensing resistors
- Dual Trak Controller
- "Bullet Proof" Function
- MIC (Enhanced IDAIN) Interface
- Test Adaptor Interface

7. **INTERNAL CLOSED WATER SYSTEM (INSIDE THE CABINET)**

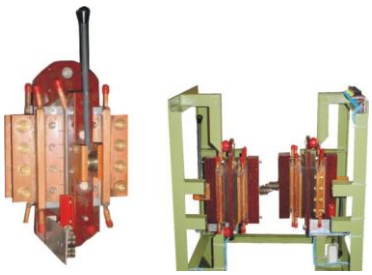
This contains one feed manifold with temperature and pressure switches and one drain manifold with temperature sensors for different paths of cooling system.

II. INTERNAL CLOSED WATER SYSTEM (Outside the cabinet)

This structure contains one plate type water to water heat exchanger, expansion/air separator tank, one mono block non-ferrous pump with starter and one deionizer cartridge for continuous purification of internal water.

III. FURNACE SELECTOR SWITCH ASSEMBLY

Furnace selector switch saves time for selecting the required furnace. For better contact it is made out of silver-plated copper blades.

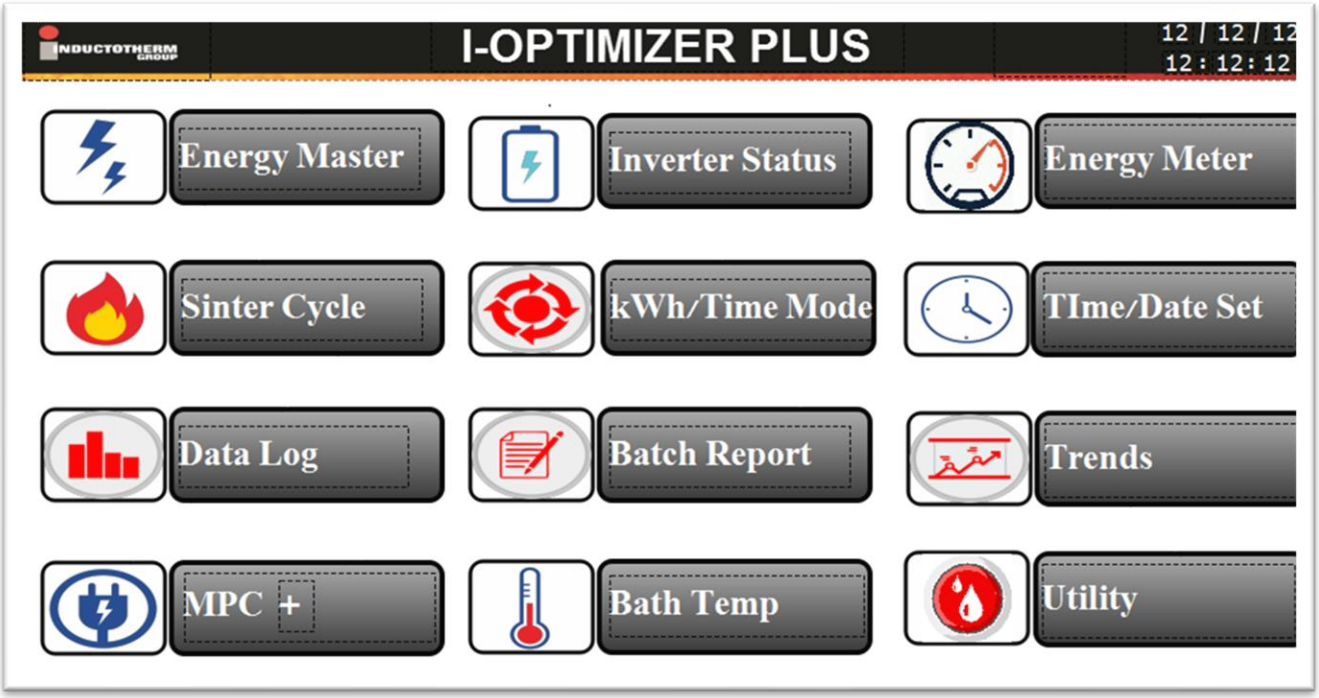


IV. HYDRAULIC POWER UNIT

One hydraulic pumping unit (without oil and starter) to supply pressurized fluid to the tilting cylinders complete with pump, pump motor, fluid reservoir, pressure relief valve, pressure gauge, return line filter and filter air breather cap all mounted on a common base with seamless pipes and fittings.

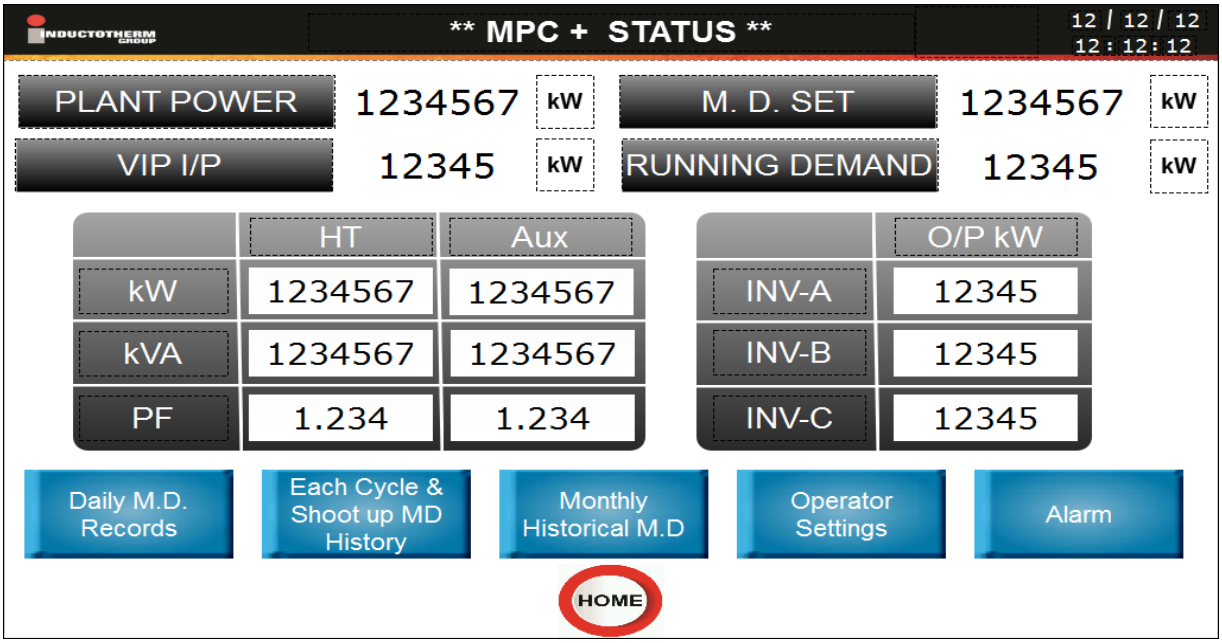
V. I OPTIMIZER plus FOR DATA ACQUISITION

For many foundries today, automation is a top priority. It is often the key to remaining competitive in what is now a challenging global castings marketplace. Inductotherm's induction power supplies combine the industry most efficient and reliable melting technology with the next generation of built-in PLC controls. All controls and system data are displayed and operated on a dedicated touch-sensitive, flat panel screen (HMI). This makes the system intuitive and easy to use.

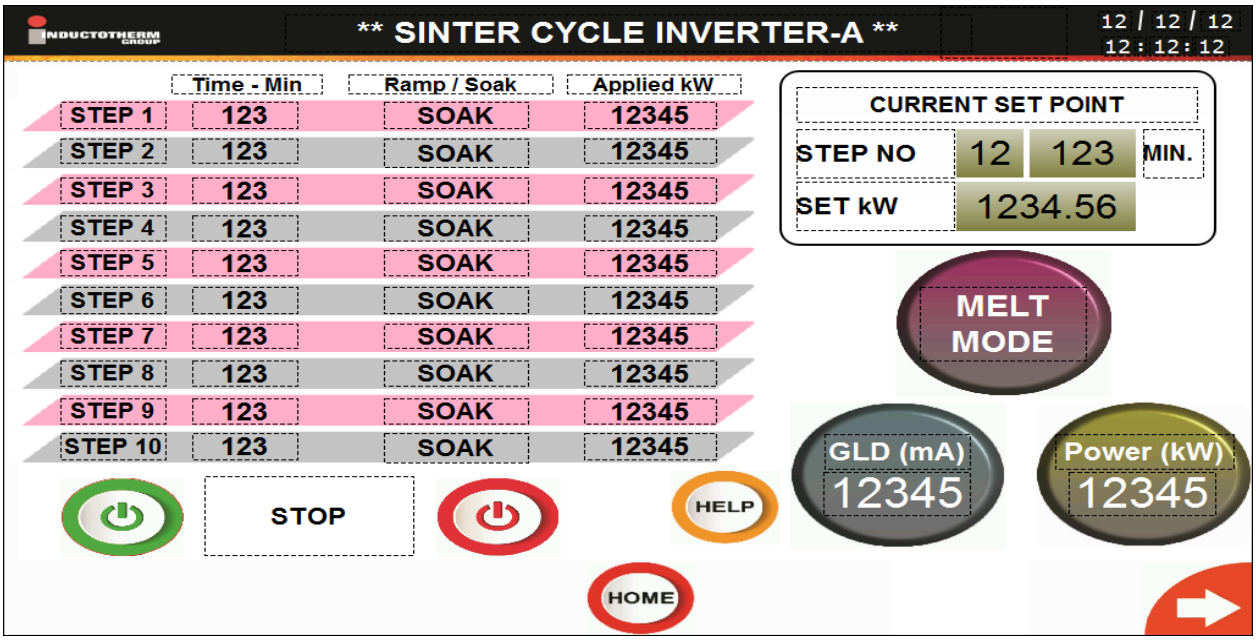


IN BUILT TOTAL PLANT MPC+ (MAXIMUM DEMAND CONTROL): Controls Plant maximum demand in every 15 minute or 30 minute cycle time as per Electricity board requirement hence reduces excess MD charges per kVA thus reduces energy cost per ton plus increases productivity by utilizing more power.

SCOPE OF SUPPLY - PAGE 5

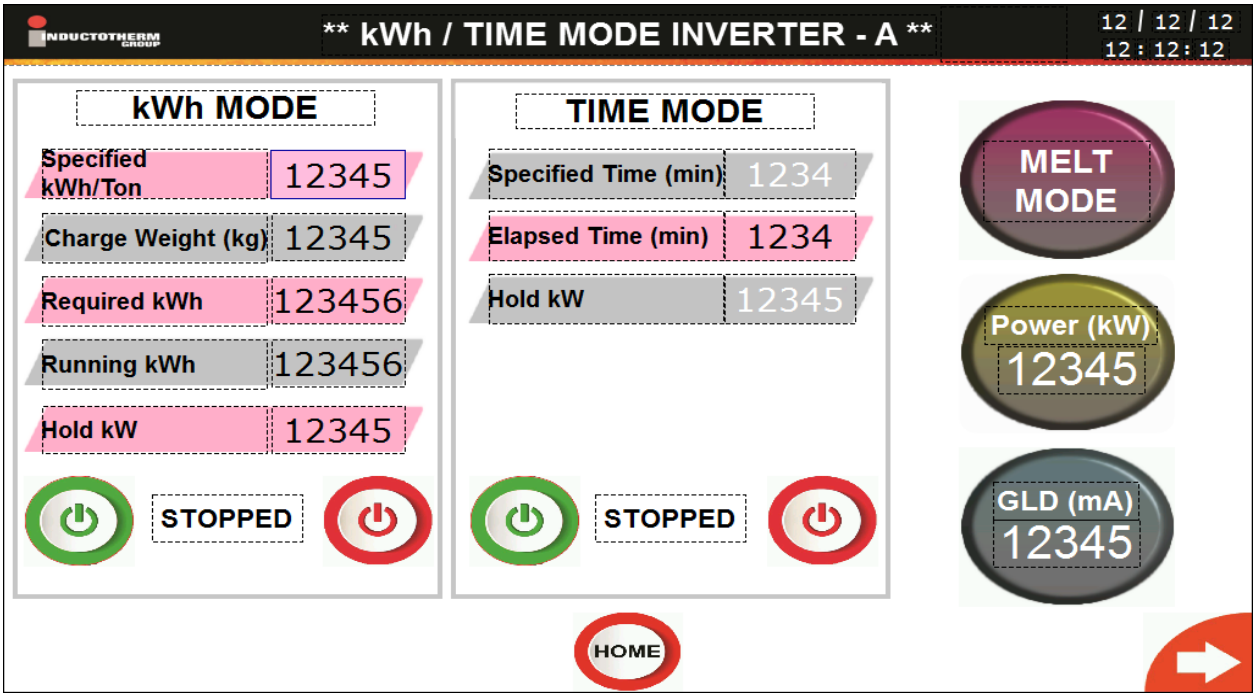


AUTO SINTERING CYCLE: Time based with 10 steps of ramp and soak with different power settings

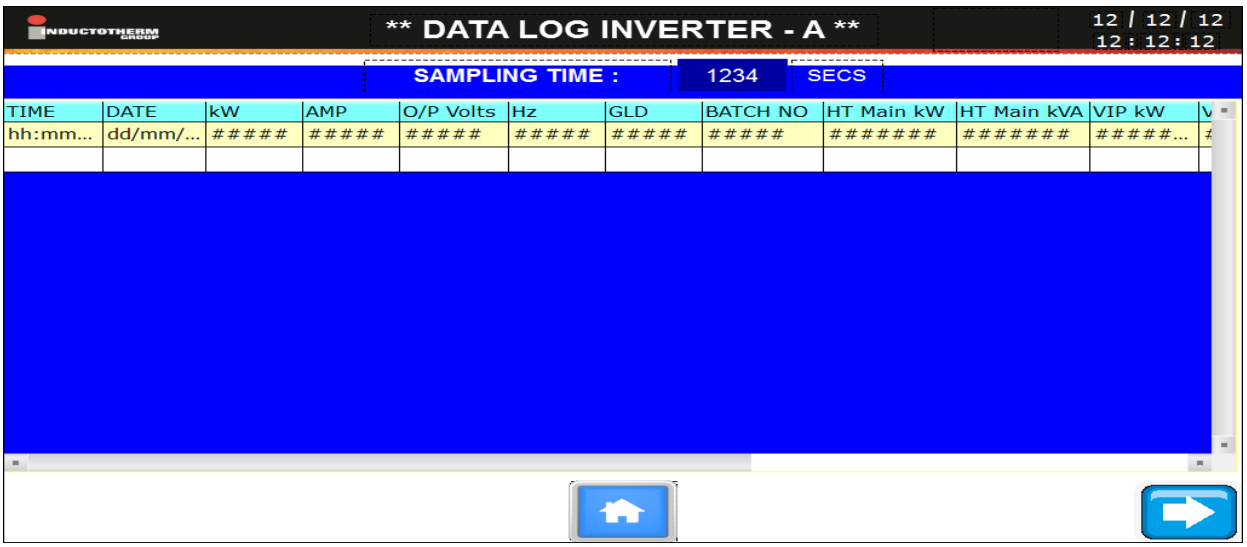


KWH MODE & TIME MODE: These are Energy Optimization modes, kWH mode keeps furnace on hold power once predefined kWH consumed similarly Time mode keeps furnace on hold power when predefined time is over, thus increase safety of the process and optimize energy consumption.

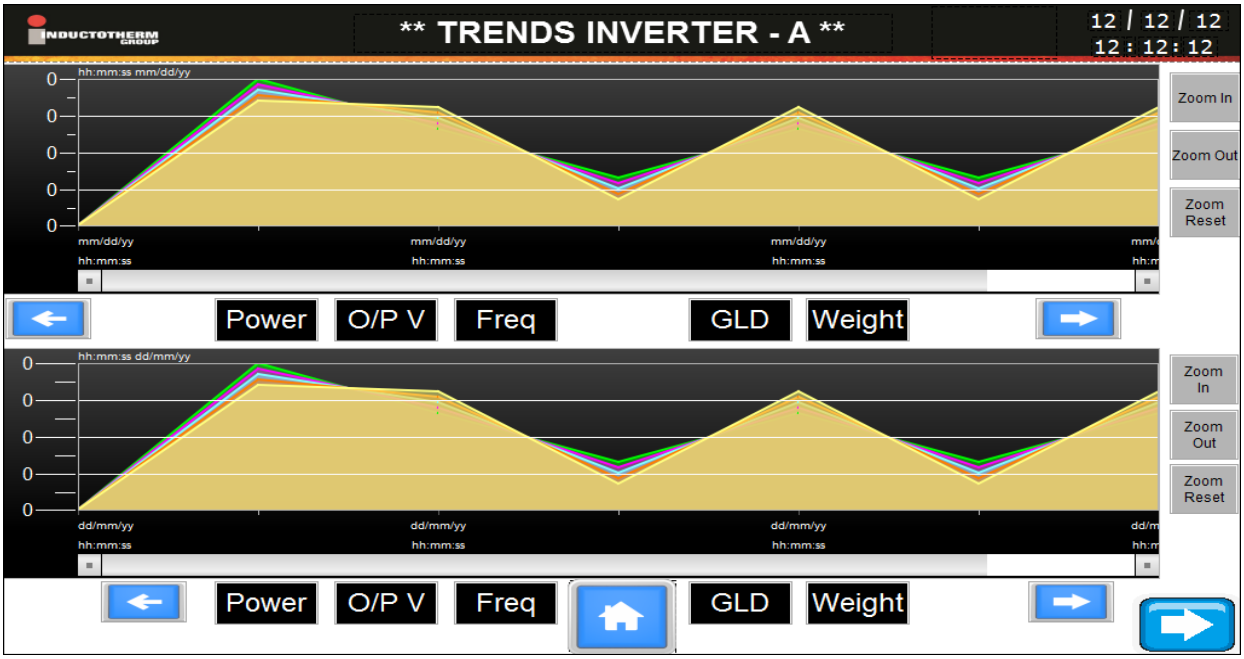
SCOPE OF SUPPLY - PAGE 6



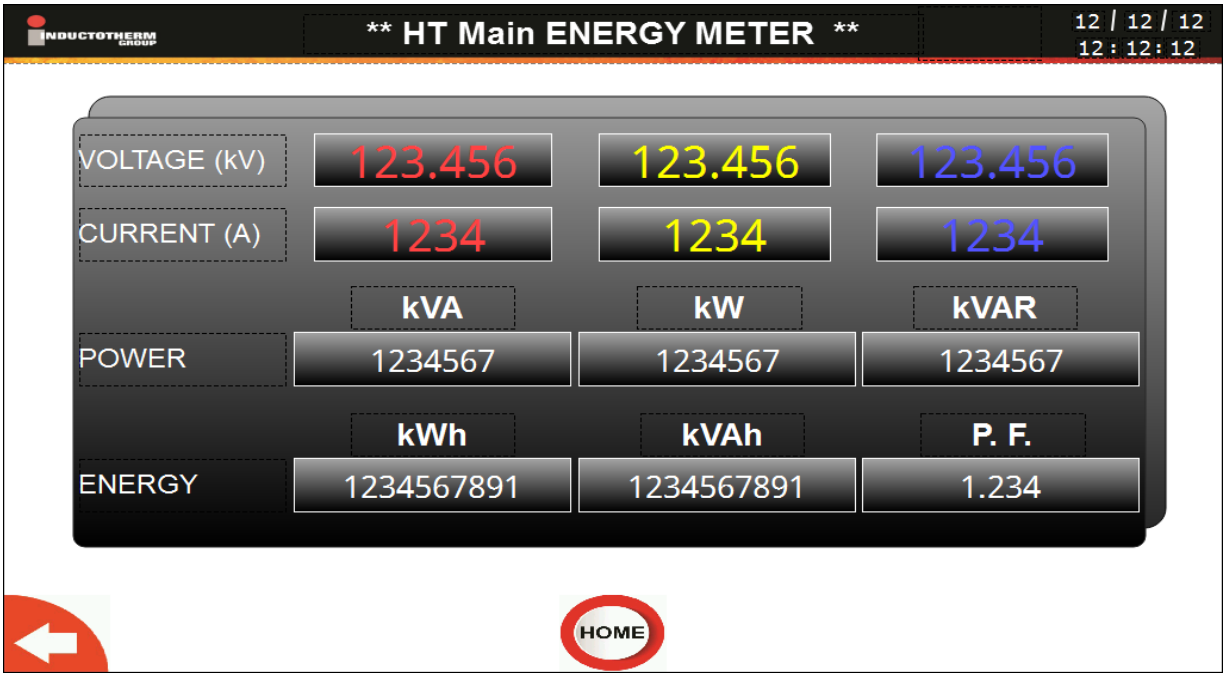
REMOTE HMI DISPLAY & EXCEL DATA LOGS & TRENDS: Due to Ethernet connectivity in PLC, customer computer can be connected remotely via Ethernet hence DELTA HMI all real time screens can be monitored , also Excel file generates per day every minute/selectable time interval with data logs and power trends .



SCOPE OF SUPPLY - PAGE 7

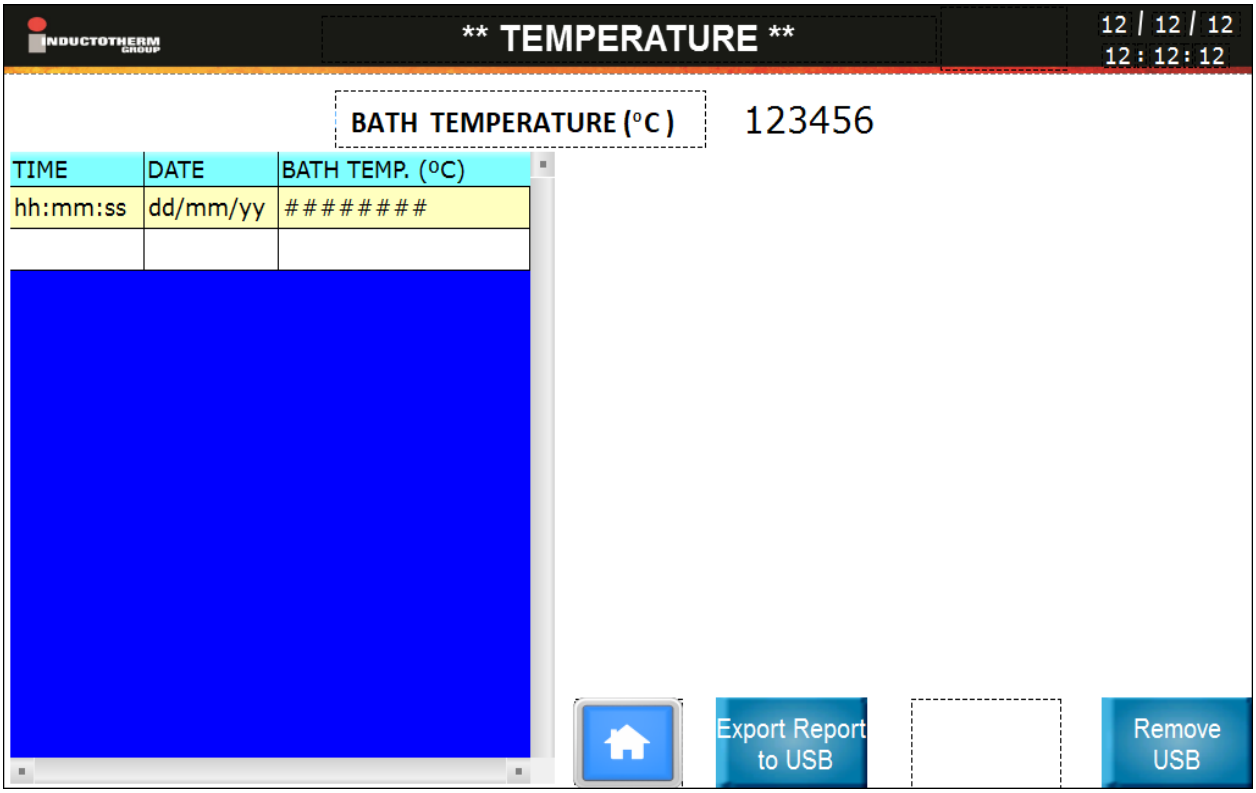


HT & INPUT ENERGY METER DATA ACQUISITION



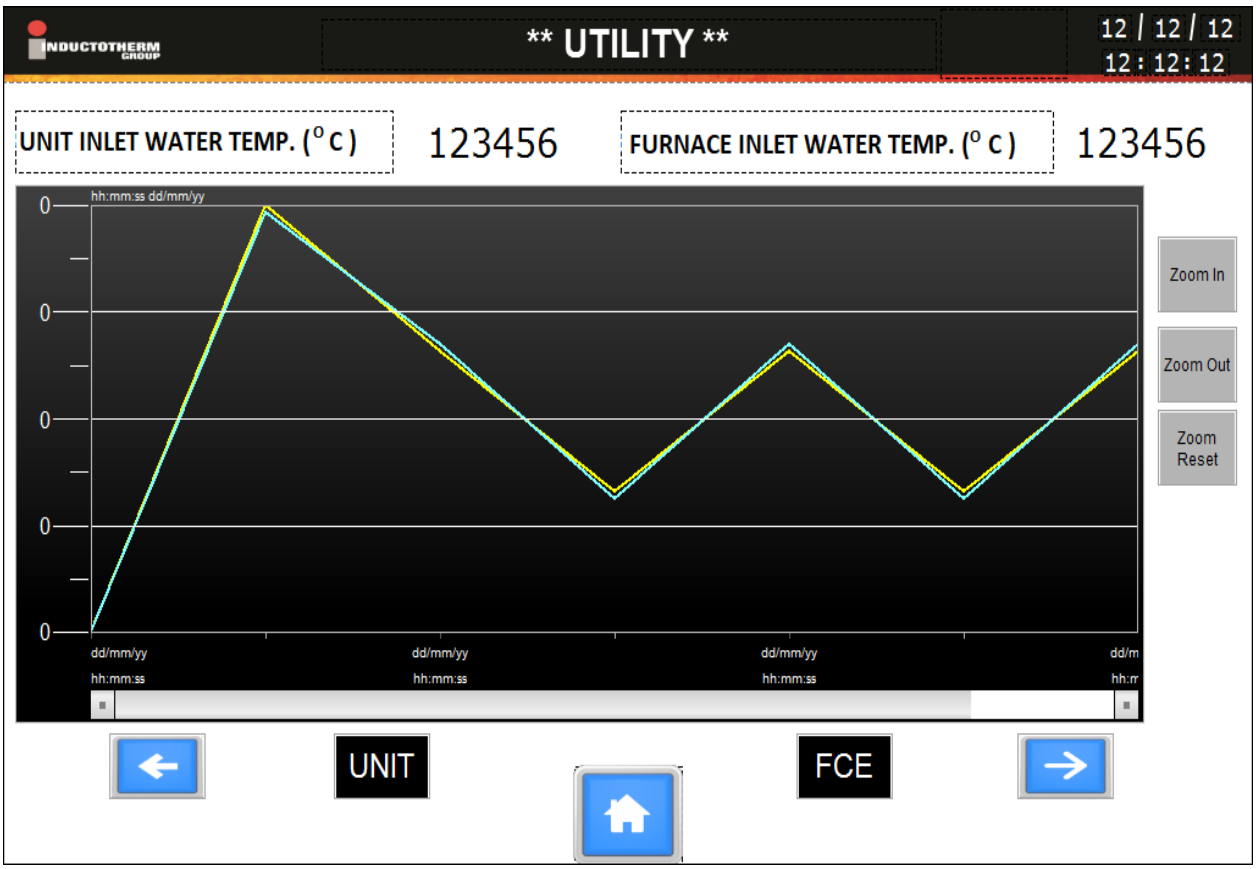
BATH TEMPERATURE

SCOPE OF SUPPLY - PAGE 8



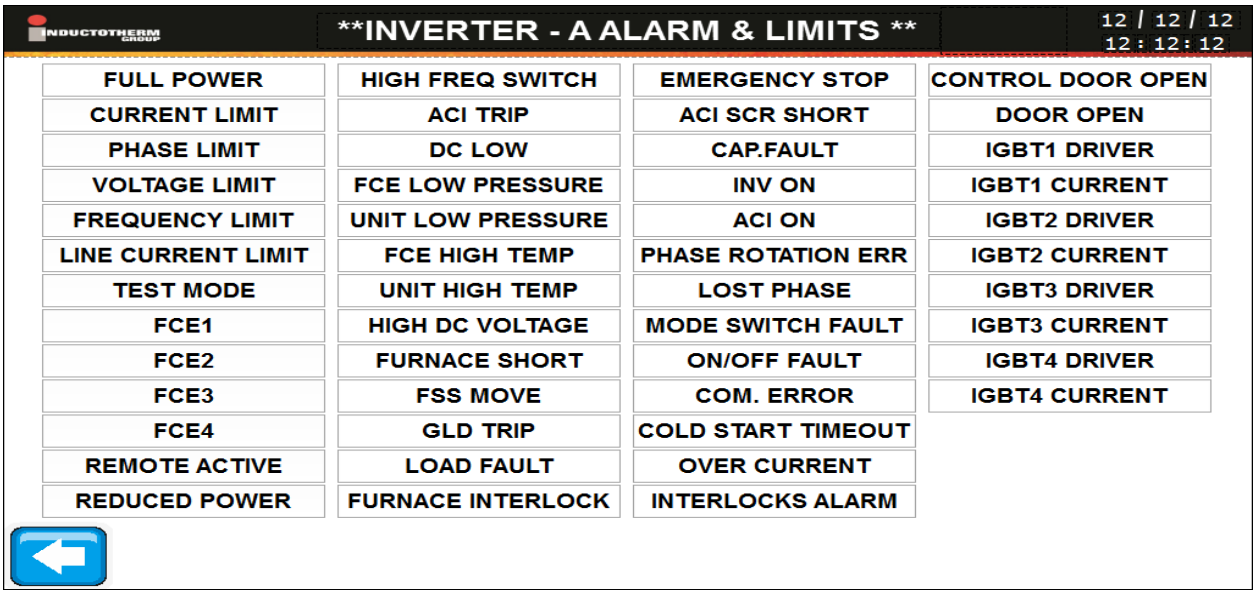
WEIGHT: Actual weight of the liquid metal will be recorded and displayed.

FURNACE INLET AND DISTILLEL WATER TEMPERATURE



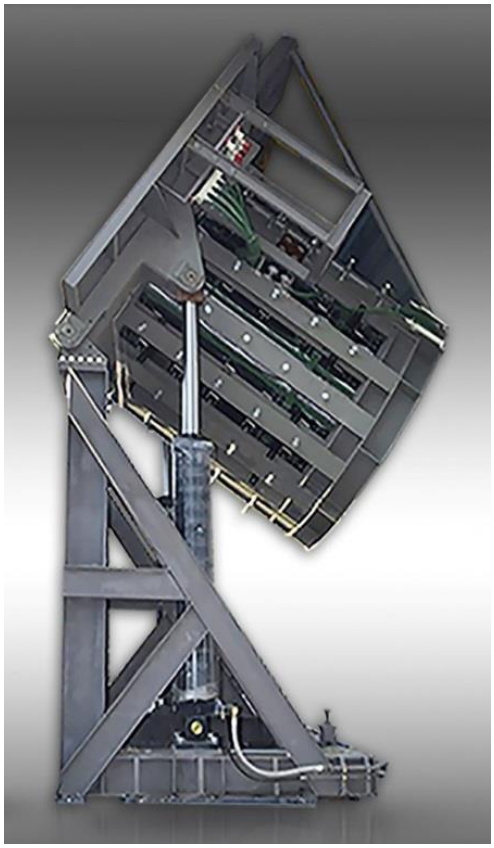
SCOPE OF SUPPLY - PAGE 9

ALARMS MONITORING & LOGGING: Safety interlocks/alarms/trip monitoring and event logging



B. MELTING FURNACE

VI. STEEL FRAME FURNACE



Hydraulically tilted steel frame furnace constructed from steel structure to provide higher rigidity and strength. Iron shunts prevent heating of the steel frame from the strong magnetic field; the shunt also holds the coil and prevents the coil from deformation.

The induction coil is placed between the special top refractory block and bottom made out of special refractory clamped by a vertical steel channel.

The coil is made out of electrolytic hollow copper section with top and bottom cooling turns made out of non-magnetic stainless steel. The complete steel frame is pivoted on strong stanchions.

Included in each furnace will be:

1. A set of hydraulic cylinders for the hydraulic tilting of the furnace.
2. Manually operated hydraulic direction control valve for tilting.
3. Leak detector assembly with stainless steel probe wires and hardware.
4. Set of flexible water-cooled power leads for connection between the power induction coil and power supply unit. Water cooled leads are with sleeves for protection against metal splash.

VII. INTERCONNECTING ARRANGEMENT

A suitable size of air/water cooled copper conductor is provided to connect the power panel with the melting furnace.

C. DRAWINGS & MANUALS

Equipment layout drawings, wiring and water diagrams, equipment outline, furnace cross section drawing and an operating and maintenance manual.

All Photographs are indicated

NECESSARY REQUIREMENTS AT CUSTOMER'S END

(CUSTOMER'S SCOPE OF SUPPLY)

Procurement and installations of following equipment and systems is customer's responsibility.

- Power line up to furnace transformer with necessary switchgears.
- 380 V, 3 Phase/60 Hz power supply from auxiliary transformer to MCC and LT cables from MCC panel to all the auxiliaries like external Pumps, Cooling tower, internal water pump and hydraulic power unit.
- Cooling water system including RCC/plastic water tanks, piping & fittings as suggested by our Project Engineering Department.
- Overhead tank for emergency water supply to coil in case of failure of water system.
- Complete civil work like furnace platform, foundations, overhead tank, underground tank, etc. Supplier will provide necessary foundation layout drawing/load data.
- Overhead crane, pouring ladles/system, pyrometer, charging device, ramming tools & other misc. equipment/tools required to start/run the system.
- Pressurized air supply if required
- Consumables like hydraulic oil, distilled water, silica paper/board, ramming mass, lining former etc.
- Statutory requirements of electricity board, factory inspectorate, pollution control and any other statutory requirements.
- 1KVA online UPS for Control System.
- Any item not specifically included in our offer

INTERNATIONAL TERMS & CONDITIONS

These terms and conditions are part of our quotation, which also contains the prices of the equipment and some of the principal terms and conditions of our offer.

EXCLUSIVE AND ENTIRE:

The following standard terms and conditions are intended by the parties to govern all the purchases of equipment, parts or service from seller, and together with the specifications provided or embodied herewith, represent the entire understanding of the parties without exception. All other terms and conditions are specifically rejected, and by proceeding with the transaction in any manner, both seller and purchaser agree that these standard terms and conditions shall control. The price charged by seller is based on these standard terms and conditions.

1. PRICE

All the prices quoted are in US Dollars, export packed **FOB MUMBAI.** This quotation is valid for Sixty (60) days only and prices should be reconfirmed after this period. The quoted prices do not include any duties & taxes.

2. DRAWINGS

Complete drawings to assist in installation and maintenance, consisting of customized layout of the equipment, standard outlines of equipment, equipment weights, electrical service, water cooling specifications; replacement parts data; and operations and maintenance manuals. Two (2) sets of drawings are provided.

3. TRAINING

The training course covers the operation and maintenance of the equipment. The course also covers component description and function of the power supply, system controls; power circuit and coils followed by operation, maintenance, troubleshooting, and shop floor safety procedures. This course is conducted thrice a year at seller's manufacturing facility in Sanand, Ahmedabad.

4. TERMS

If Applicable, confirmed irrevocable Letter of Credit drawn on HDFC Bank Limited, Navrangpura Branch, Ahmedabad , Gujarat, India, acceptable to Inductotherm (India) Pvt. Ltd, to be established at the time of placement of order and the L.C. to be kept valid for a period not less than 6 weeks after promised delivery and payable upon presentation of invoice and on-board bills of lading. It will be the Purchaser's responsibility to extend the validity of the L/C to accommodate any delays in delivery due to export/import license requirement

5. INSURANCE

Purchaser represents that they have a program of Insurance which adequately protects their interest, and that of their employees, including damage to plant, property and equipment, personal injury of any kind, directly or indirectly related in any way to the equipment, service, repair or parts supplied by Seller. Accordingly, Purchaser waives any claim against seller for the foregoing, and on behalf of its Insurance company, any right of subrogation in connection therewith.

6. CHANGE IN ORDER

This may be accommodated at the sole discretion of seller, but the original order will be treated as cancelled and cancellation charges may be applicable. The revised order will have to be at the then prevailing terms and conditions of prices and delivery.

7. TERMINATION

As Induction furnace is capital equipment, it is mostly made against an order. In the event that this purchase is terminated for any reason (including Purchaser's failure to make partial payments) engineering efforts and material will result in loss to Seller. To compensate this loss, the advance will not be refundable and in case if expenses incurred towards engineering efforts and material exceeds advance collected, the balance amount will be payable by the purchaser.

8. DELIVERY

Normal delivery on the above type of equipment is from three (3) to Eight (8) months following receipt of technically and commercially clear order. However, Inductotherm (India) Pvt. Ltd. will, by policy, always endeavor to meet deliveries as required by our purchasers subject to availability of components/ parts from our suppliers. Actual delivery of the equipment covered by this quotation will be set during final negotiations and upon sufficient clarification of commercial details and final specifications to allow project to proceed.

9. INSTALLATION

The supplier's project engineering personnel will guide purchaser's technical people for installation of equipment, selection and location of auxiliary equipment's like cooling tower, transformer, pipes, pumps, heat exchanger, etc. if applicable. They will not be available on a day-to-day basis to supervise erection job. Customer will have to provide for his to & fro air travel, visa expenses, living, boarding, internal travelling and local conveyance expenses.

10. STARTUP

The normal period of start-up of the equipment is about three to five days, if all the site material and necessary arrangements are complete. The services of a qualified field engineer to supervise proper start-up and commissioning of the equipment will be chargeable as USD 300/- per day. Customer will have to provide for his to & fro air travel, visa expenses, living, boarding, internal travelling and local conveyance expenses The purchaser will be charged extra for additional man-days at the prevailing service charges. Excess material, if any, left after commissioning is Inductotherm's property and will be taken back or can be retained by purchaser on chargeable basis.

11. WARRANTY

The warranty stated here is valid only if supplier's engineer has commissioned the equipment. Supplier agrees to repair or replace ex-factory parts manufactured by supplier except expendable items such as refractories, refractory components, fuses, bulbs, pumps seals, melting crucibles, thermocouples, water cooled leads, Hydraulic hose pipes and other similar consumable items that fail due to defects in material or workmanship within one year of start-up or eighteen (18) months after the shipment, whichever occurs first. The warranty obligations of Seller with respect to equipment not manufactured by Seller shall conform to and be limited to the warranty actually extended to Seller by its suppliers. Notice of a claim for alleged defective equipment must be given within fifteen (15) days after Purchaser learns of the defect. The defective part or parts shall be returned to Seller, freight prepaid, unless otherwise directed by Seller. This warranty shall be exclusive and in lieu of any other warranties and seller makes no warranty of merchantability or warranties of any other kind express or implied, including any implied warranty of fitness for a particular purpose which extend beyond the warranty as set forth above. Seller's liability for any and all losses and damages to purchaser resulting from defective parts of equipment shall in no event exceed the cost of repair or replacement, EX-WORKS of defective parts or equipment. IN NO EVENT SHALL SELLER BE LIABLE FOR INCIDENTAL, SPECIAL OR CONSEQUENTIAL DAMAGES OF ANY KIND OR NATURE WHATSOEVER OR LOST PROFITS.

Please note that if cause of failure is identified as arising out of improper operation, maintenance negligence, tampering or any accident etc. then such parts and labour are not covered under above warranty clause. This will be judged fairly by a team of our technical experts at Ahmedabad in consultation with the concerned field engineers. Furthermore, seller is not responsible for any damage resulting from chemical/or metallurgical reaction.
The warranty will lapse if the purchaser uses components from any other suppliers/manufacturers without supplier's written permission Or if servicing is done by other than Inductotherm engineers; or sale of the equipment within warranty period.

12. AFTER SALES SERVICE

During the warranty period, if a fault occurs customer will have to provide for service engineer to & fro air travel, visa expenses, living, boarding, internal travelling and local conveyance expenses of travelling and living expenses. No daily service charge will be billed. However, after warranty period, daily service charges as per our prevailing service charges per engineer shall be applicable along with to & fro air travel, visa expenses, living, boarding, internal travelling and local conveyance expenses of travelling and living expenses. Supplier's service engineer will attend faults, material required for rectifying the equipment will be provided by purchaser.

13. LAW

This agreement shall be governed by the internal law of the state of Gujarat, India, and any claims arising hereunder shall be prosecuted in the India District Court having jurisdiction of the cause of action arising in the district in which Inductotherm is located.

14. PATENTS

Seller agrees to indemnify Purchaser against any proven claim and assessed liability for infringement of any United States patent arising from the manufacture or sale of any apparatus furnished by Seller to Purchaser. THE FOREGOING STATES SELLER'S ENTIRE LIABILITY FOR CLAIMS OR PATENT INFRINGEMENT. Seller shall have no liability whatsoever if the claim of infringement arises out of Seller's compliance with Purchaser's specifications. Seller shall have no liability whatsoever if a claim of infringement is based upon the Purchaser's use of the equipment as part of a patented combination where the other elements of the combination are not supplied by Seller, or in the practice of a patented process. Where the specifications, process, design are supplied by Purchaser, then Purchaser agrees to indemnify Seller in like manner.

15. PROPRIETARY RIGHTS

Unless otherwise stated herein all design, manufacturing processes, manufacturing information, vendor sources, know-how, equipment, tooling or other hardware, software, or information (collectively referred to as "resources") acquired or utilized by the Seller to produce the finished goods, and any intellectual property rights, including but not limited to patents, copyrights and trade secrets related in any way to the resources, are and shall hereinafter remain the exclusive property of the Seller, regardless of whether such resources are created solely by the Seller, or by Purchaser's collaboration with the Seller, for example, where Seller utilizes Purchaser's specifications to create resources, and the Purchaser shall acquire or receive no rights or title therein or thereto as a result of this purchase whether or not the order provides for Seller's delivery of technical data, drawings or other information to the Purchaser in addition to the finished goods. Unless Seller's prior written consent is given, in no event shall the Purchaser permit such data, drawings or information to be: (1) disclosed to any third party other than the Purchaser's customer; (2) used by the Purchaser or the Purchaser's customer for manufacture of like or similar goods; (3) used for purposes of duplicating or reverse-engineering Seller's proprietary designs or processes; or (4) used by a party other than the Purchaser or the Purchaser's customer for any purpose.

16. U.S. EXPORT CONTROLS

All items furnished by Seller to Purchaser in connection herewith shall at all times be subject to the export control laws and regulations of the U.S. including, but not limited to, 10 CFR Part 810 and U.S. Export Administration Regulations. Purchaser agrees and give assurance that no items, equipment, materials, service, technical data, technology, software or other technical information or assistance furnished by Seller, or any good or product resulting therefrom, shall be exported or re-exported by Purchaser or its authorized transferees, if any, directly or indirectly, unless in accordance with applicable U.S. export laws and regulations. The aforesaid obligations shall survive any satisfaction, expiration, termination or discharge of any other contract obligations. Unless otherwise specified, purchaser will not use products directly or indirectly, in any application involving missile technology, nuclear proliferation, or chemical or biological weapons proliferation.

17. LIMITATION OF LIABILITY

Seller's total aggregate liability (arising out of or in connection with but not limited to any breach of contract, negligence, tort, liquidated damages, specific performance, termination, cancellation including the repaying of the contract price or parts thereof, fundamental breach of warranties, misrepresentation, nonperformance, nonpayment or any other) whether based in contract, in tort, in equity, on statute, at law or on any other theory of law, shall not exceed the paid contract price. The purchaser acknowledges that the remedies provided in this contract are exclusive and in lieu of all other remedies available to the purchaser at law, in contract, in tort, in statute or in equity or in any other theory of laws.

18. SEVERABLE AND INDEPENDENT PROVISION

Without limiting the foregoing, it is expressly understood and agreed that each and every provision of this agreement that provides for a limitation of liability, disclaimer of warranties, or exclusion of damages is intended by the parties to be severable and independent of any other provision and to be enforced as such. Further, it is expressly understood and agreed that in the event any remedy hereunder is determined to have failed its essential purpose, all other limitations of liability and exclusion of damages set forth herein shall remain in full force and effect. All such provisions shall survive the termination of this agreement.

19. Business Practice:

Purchaser and Seller shall comply with applicable anti-corruption laws and regulations. Specifically, Purchaser and Seller warrant that neither it nor any of its employees, agents, or representatives have offered or given, or will offer or give, or authorize to give, any payments, gifts, property, or thing of value, directly or indirectly, to any person, official, employee or representative of any government or agency or instrumentality thereof, or to any employee, agent, or representative of the other Party, for the purpose of securing any business or influencing such person with respect to the terms, conditions, or performance of any contract, including this order. If Purchaser breaches the terms of this section, Seller may immediately terminate this order without any liability.

20. FORCE MAJURE

If the performance of any part of this contract by seller is rendered commercially impracticable by reasons of any strike, fire, flood, accident, or any other contingency, the non-occurrence of which was a basic assumption of this contract including war, embargo, government regulation, or any unforeseen shutdown of major supply or other like causes beyond the control of seller such as cannot be circumvented by reasonable diligence and without unusual expense, Seller shall be excused from such performance in whole or in part to the extent that it is prevented and for as long a period of time as these conditions render seller's performance commercially impractical.

For, INDUCTOTHERM (INDIA) PVT. LTD.

AUTHORISED SIGNATORY